No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	**A Comprehensive Measure of Well-Being**

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Dataset Loading	Loads the HDI dataset from a CSV file for preprocessing and training.	Data Processing	Done	
2	Data Processing	Cleans the dataset, handles missing values, and performs label encoding where required.	Data Processing	Done	
3	HDI Prediction Model	Trains a Linear Regression model to predict the Human Development Index.	Machine Learning	Done	
4	Model Serialization	Saves the trained model using Pickle for future predictions.	Model Management	Done	
5	Flask Web Application	Provides a web interface for users to enter input values and request predictions.	Web Application	Done	
6	Prediction Result Display	Displays the predicted HDI value and Human Development category on the results page.	User Interface	Done	
7	Data Visualisation	Generates graphs and charts for dataset analysis using Matplotlib and Seaborn.	Data Visualisation	Done	
8	Online Deployment	Deploys the Flask application on Render for public access.	Deployment	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Flask web pages, Prediction Result Page
2	Backend / Logic	3	Data preprocessing, Linear Regression model, Prediction engine
3	Database / Storage	2	HDI.csv dataset, Pickle (model.pkl)
4	API / Integration	1	Flask routing between frontend and prediction model
5	Security / Authentication	0	Flask routing between frontend and prediction model